Cross-Domain Relation Classification: A Critical Evaluation of Different Domain-Information Approaches

Second Year Project (Introduction to Natural Language Processing and Deep Learning)

BSSEYEP1KU

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Abstract

With the rise in popularity of using artificial intelligence for natural language based tasks, certain challenges arise. We focus specifically on the task of relation classification, which is a very domain-specific task. It is common to want a model that works across multiple domains, but current methods of relation classification do not perform well in cross-domain cases. This means having to train a separate model for each domain to get the best performance.

To reduce the increasing cost of training large language models, we aim to provide a method for training a model for cross-domain relation classification on multiple domains. We propose two methods, one using a special token to encode the domain, and another using dataset embeddings to encode the domain.

Our results show that the model using special domain tokens is able to marginally outperform the baseline case, however the dataset embeddings significantly worsen the model performance.

1 Introduction

Relation Classification (RC) is the task of identifying the relationship between two entities in a given sentence. RC is an important step in Relation Extraction (RE), typically used for extracting specific information from a body of text. These processes are cornerstones in different Natural Language Processing (NLP) applications which handle tasks such as summarising text, question answering (e.g., chatbots), populating knowledge bases, etc.

The task of RC has been widely researched, using different approaches such as traditional neural models, CNN-based models and RNN-based models (Miwa and Bansal, 2016). However the focus of model evaluation is mainly limited to in-domain setups. In general, there is a gap in the research of cross-domain RE, implying a gap in the case of cross-domain RC as well (Bassignana and Plank, 2022).

We seek to explore the possible benefits of adding additional information about the given domain to the input sentences. We explore two different approaches; the first involves adding a special token to the beginning of the sentence containing the domain, and the other approach makes use of dataset embeddings. These embeddings are summed to the original input embedding of the model, and are learned during training, as described by van der Goot and de Lhoneux (2021).

This leads us to the following research-question: *Is it possible to improve the performance of multi-domain RC by enriching the input data with information about which domain a sentence is from?*

2 Related Work

In the case of cross-domain NLP tasks, adding similarities between datasets is useful, but this needs to be done while still holding on to the domain-specific information. RC is domain specific, and the performance of the model is highly dependant on how much the test data looks like the domain the model has been trained on.

Studies show that using shared representations of words across languages offers intriguing possibilities. Using vector representations of words that are shared across languages, it is possible, for example in machine translation, for a model to translate a word that it has never been seen before. The model can find a similar word in the target language based on the embeddings of the two words, which are automatically learned during training (Ammar et al., 2016).

Other domain information methods have also been studied by Alumäe (2013), where the proposed model also trains a vector of factors for each domain, which are then used in the transition between the projection layer and the hidden layer.

Dataset embeddings, also referred to as lan-

guage embeddings, is yet another approach that allows the model to learn general features across domains while also distinguishing certain features between domains (van der Goot and de Lhoneux, 2021). Their experiments show positive results using dataset embeddings for dependency parsing. Therefore we are hopeful that it will also be able to improve performance in relation classification.

3 Data

We use the CROSSRE dataset, a dataset targeting cross-domain evaluation for RE for English (Bassignana and Plank, 2022). The dataset consists of six different domains: news (), politics (), natural science (), music (,) literature () and artificial intelligence (;AI). The news domain is collected from Reuters News and the remaining five domains are collected from Wikipedia. They all share the same 17 relation labels. The train/dev/test data split can be seen in Table 1. We can see that there is a fairly similar amount of sentences and relations in most domains, with the exception of the news domain, which has much fewer relations. This could have an effect on how well the model learns that domain compared to the other ones.

	SENTENCES				RELATIONS			
	train	dev	test	**tot.**	train	dev	test	**tot.**
	164	350	400	914	175	300	396	871
	101	350	400	851	502	1,616	1,831	3,949
	103	351	400	854	355	1,340	1,393	3,088
	100	350	399	849	496	1,861	2,333	4,690
	100	400	416	916	397	1,539	1,591	3,527
	100	350	431	881	350	1,006	1,127	2,483
tot.	668	2,151	2,446	**5,265**	2,275	7,662	8,671	**18,608**

Table 1: Train/dev/test statistics. Reprinted from Bassignana and Plank (2022)

4 Experimental Setup

In this setup we explore two different methods of encoding information about which domain each sentence comes from. Namely using special domain tokens and dataset embeddings.

For both methods as well as the baseline, we train a separate model following the state-of-the-art by Baldini Soares et al. (2019). The models all use the hyperparameters proposed by Bassignana and Plank (2022) which ensures that the performance difference is only due to the domain information and not due to better hyperparameters. For reproducibility, the code is available on GitHub [1].

Entity marked sentences are fed into a pre-trained BERT encoder, and the output representations are then used for the relation type classification via a feed-forward Neural Network (Bassignana and Plank, 2022).

Before feeding the data to our models we convert the entity labels from the entity markers into five generic labels (person, location, organisation, misc and event) that are present in all domains. This is done because some of the entity labels are only present in some domains, meaning that they could indirectly be used to identify the domain. Additionally, some labels have very sparse distributions. Converting them into fewer generic labels means that the distribution of the entity markers is more dense, leading to the model learning a more accurate representation of them.

Baseline As a baseline we trained a model on all the domains in the CROSSRE data set without any information about which domain each sentence is from, and evaluated the model on each domain separately.

Special Tokens The special token model tries to encode information about the domain by using a special token such as `[MUSIC]` or `[NEWS]` before the sentence but after the special (classification) token `[CLS]` which is needed for BERT. The token is added to the model as a special token that is not split by the tokenizer. The reasoning for putting the token uppercase inside brackets is that it will not be seen in any sentences, so the meaning of the token is learnt separately.

Dataset Embeddings The dataset embedding model tries to encode information about the domain by adding dataset embeddings on the encoder side.

Dataset embeddings are vector representations of the used datasets learned during model training that aim to capture distinctive properties of multiple data sources into a continuous vector, without losing their heterogeneous characteristics. In this case, we are treating each domain as a separate data source. Each input sentence is marked with the source domain mapped to an integer. Then for each wordpiece in the sentence, the domain integer is used to look up an embedding of size 768, which is summed with the word, segment, and position embeddings (See Figure 1). These domain specific embeddings are learned during training and

[1] https://github.com/viggo-gascou/second-year-project

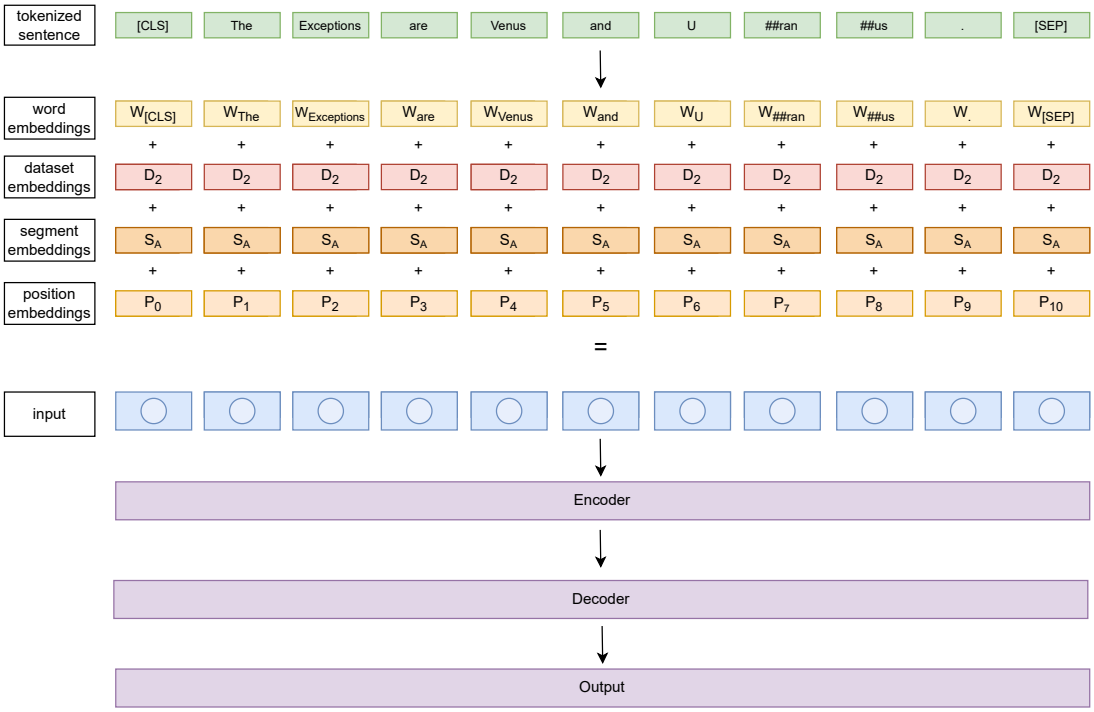

Figure 1: A visualisation of the integration of dataset embeddings with the BERT-based model

allow the model to learn useful similarities between the different domains while still encoding domain-specific knowledge (van der Goot and de Lhoneux, 2021).

5 Results

The results are presented in Table 2. The results are reported for both proposed models together with the baseline model, and have all been evaluated on each domain separately, so we can compare the performance between each domain.

We compute the following three metrics: micro-averaged F1, macro-averaged F1 and the weighted F1, in order to capture a detailed look at the performance of the models. We expect a lower macro-F1 score for all domains given the small number of per-label instances in each domain. The macro-F1 provides a better insight on the per-class performance, given that we have an imbalanced class distribution. On the contrary, the micro-F1 gives equal weight to each sample, and is a better method for a study of each individual class.

Looking at the average F1 scores, we can see that the baseline and special token models perform comparably. The baseline gets an average micro-F1 of 60.39, with special tokens scoring 60.08, so in that metric the baseline performs marginally better. They both perform vastly better than the dataset embedding model with a score of 30.99. But in macro-F1 the special tokens model gets an average score of 34.90, outperforming the baseline's score of 33.45. The dataset embedding model performs extremely poorly in this metric with an average score of 4.12. The special token model gets an average weighted F1 score of 56.65, which again performs marginally better than the baseline with a score of 56.54. In this metric, the dataset embedding model also performs very poorly with an average score of 17.58.

Curiously, the dataset embedding model performs comparably or slightly better than the other two specifically in the news domain. It gets a slightly higher micro-F1 score than the other two models, and a better weighted F1 score than the baseline. This is the only domain in which it performs anywhere close to as well as the other models. We are unsure why this is, but it certainly could be related to the much lower number of relations in the news domain compared to the other domains.

It is clear after looking at the results that neither adding special tokens nor adding dataset embeddings helps the model perform particularly better than the baseline.

								avg.
MICRO-F1	BASELINE	51.67	70.90	54.75	61.36	47.46	76.21	60.39
	SPECIAL TOKENS	49.97	70.60	54.14	61.84	47.74	76.17	60.08
	DATASET EMBEDDINGS	18.69	19.03	58.54	42.23	17.03	30.43	30.99
MACRO-F1	BASELINE	39.53	42.10	14.39	27.71	38.21	38.77	33.45
	SPECIAL TOKENS	39.67	42.65	16.06	30.19	40.34	40.50	34.90
	DATASET EMBEDDINGS	1.88	1.90	12.73	3.68	1.72	2.78	4.12
WEIGH.-F1	BASELINE	48.81	66.08	48.97	57.55	45.48	72.35	56.54
	SPECIAL TOKENS	47.43	66.45	49.30	58.44	45.48	72.79	56.65
	DATASET EMBEDDINGS	5.87	6.08	49.55	24.63	5.20	14.14	17.58

Table 2: Experiment test results, averaged over the 5 seeds.

5.1 Evaluation

5.1.1 Significance Testing

The conclusions we draw from the results are based on the Almost Stochastic Order test (ASO) (Del Barrio et al., 2018; Dror et al., 2019) implemented by Ulmer et al. (2022). We compare the baseline and the special tokens model by running the test with a confidence level of 95% adjusted with the Bonferroni correction. We found the distribution of the scores based on five random seeds to not be stochastically dominant over the baseline model. We also compare the model using dataset embeddings to the baseline with the same approach as above. Similar results were found. Overall, there is no statistical evidence that any of the two models perform better than the baseline. (See Appendix B for ASO scores).

5.1.2 Analysis of embeddings

To help determine if the dataset embeddings actually encoded meaning about the domains, we extracted the embeddings and reduced the number of dimensions using principal component analysis (PCA). Analysing the embeddings this way has also been done in previous work (van der Goot et al., 2021).

From the plot of the first two principal components (Figure 2) we see that the news and politics domains are more related to each other, as they are relatively close to each other in the projected space. This is also what we would intuitively expect. The science and music domains are far from each other, which we would expect as those subjects have little in common with each other.

We did the same procedure with the embeddings for the special tokens. Here we also see the news and politics domains close to each other. The music and science domains far from the other domains, which we would also expect. Clearly it seems some sort of information about the domains and their relation to each other seems to be learned in both the dataset embeddings and special token embeddings.

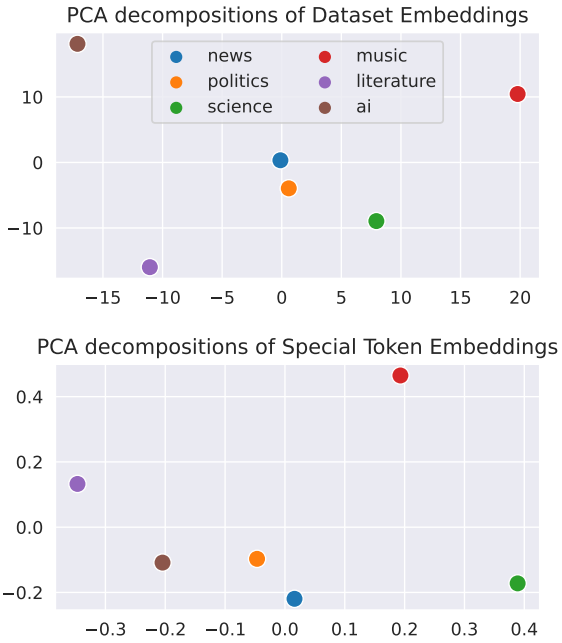

Figure 2: The first two principal components of the domain dataset embeddings

6 Conclusion

In this paper, we explored two approaches of adding domain knowledge to improve performance of cross-domain relation classification.

The special tokens model includes domain information by adding a domain specific token to the input sentence. The other approach uses dataset embeddings, an additional layer of embeddings added to the word-embeddings, which are learned

during training.

From the results of this experiments, we were not able to improve the performance of multi-domain RC by enriching the input data with information about which domain a sentence is from. However, we cannot reject that it may be possible, and further research could be done to test other methods or fine tune the special token approach, which performs barely better than the baseline.

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A Per-class results

BASELINE	RELATED-TO	0.41	0.10	0.00	0.05	0.40	0.07
	ARTIFACT	0.65	0.89	-	0.36	0.59	0.86
	CAUSE-EFFECT	0.00	0.00	-	-	0.00	0.00
	COMPARE	0.03	0.00	-	0.00	0.12	0.00
	GENERAL-AFFILIATION	0.14	0.83	0.32	0.36	0.44	0.85
	NAMED	0.79	0.61	0.41	0.76	0.65	0.45
	OPPOSITE	0.09	0.00	0.09	0.22	0.00	0.00
	ORIGIN	0.54	0.28	-	0.20	0.43	0.30
	PART-OF	0.51	0.15	0.03	0.19	0.28	0.29
	PHYSICAL	0.85	0.82	0.71	0.78	0.79	0.94
	ROLE	0.69	0.64	0.64	0.56	0.48	0.64
	SOCIAL	0.00	0.48	-	0.24	0.28	0.07
	TEMPORAL	0.70	0.74	0.24	0.87	0.63	0.50
	TYPE-OF	0.24	0.76	-	0.04	0.09	0.74
	USAGE	0.11	0.00	-	-	0.00	0.07
	WIN-DEFEAT	0.75	0.75	0.00	0.08	0.84	0.81
SPECIAL TOKENS	RELATED-TO	0.37	0.17	0.18	0.07	0.43	0.13
	ARTIFACT	0.60	0.87	-	0.43	0.59	0.86
	CAUSE-EFFECT	0.00	0.00	-	-	0.00	0.00
	COMPARE	0.01	0.00	-	0.00	0.16	0.00
	GENERAL-AFFILIATION	0.18	0.84	0.25	0.35	0.48	0.86
	NAMED	0.76	0.62	0.39	0.74	0.63	0.50
	OPPOSITE	0.06	0.09	0.10	0.27	0.00	0.05
	ORIGIN	0.53	0.34	-	0.20	0.42	0.34
	PART-OF	0.50	0.18	0.11	0.25	0.27	0.31
	PHYSICAL	0.87	0.82	0.69	0.78	0.80	0.95
	ROLE	0.66	0.64	0.65	0.57	0.41	0.63
	SOCIAL	0.20	0.51	-	0.42	0.43	0.23
	TEMPORAL	0.72	0.76	0.35	0.86	0.68	0.50
	TYPE-OF	0.22	0.62	-	0.00	0.10	0.72
	USAGE	0.11	0.00	-	-	0.00	0.01
	WIN-DEFEAT	0.77	0.74	0.01	0.13	0.84	0.77
DATASET EMBEDDINGS	RELATED-TO	0.00	0.00	0.03	0.00	0.03	0.00
	ARTIFACT	0.00	0.25	-	0.00	0.00	0.00
	CAUSE-EFFECT	0.00	0.00	-	-	0.00	0.00
	COMPARE	0.00	0.00	-	0.00	0.00	0.00
	GENERAL-AFFILIATION	0.00	0.07	0.00	0.00	0.00	0.47
	NAMED	0.00	0.00	0.64	0.00	0.00	0.00
	OPPOSITE	0.00	0.00	0.11	0.00	0.00	0.00
	ORIGIN	0.00	0.00	-	0.00	0.00	0.00
	PART-OF	0.32	0.00	0.00	0.00	0.00	0.00
	PHYSICAL	0.00	0.00	0.67	0.00	0.06	0.00
	ROLE	0.00	0.00	0.71	0.63	0.20	0.00
	SOCIAL	0.00	0.00	-	0.00	0.00	0.00
	TEMPORAL	0.00	0.00	0.00	0.00	0.00	0.00
	TYPE-OF	0.00	0.00	-	0.00	0.00	0.00
	USAGE	0.00	0.00	-	-	0.00	0.00
	WIN-DEFEAT	0.00	0.00	0.00	0.00	0.00	0.00

Table 3: Per-class F1-scores for each model. Averaged over the 5 seeds. '-' indicates the class is not present in the test set.

B ASO Scores comparing our models and the baseline

SEED	4012	5096	8878	8857	9908
SPECIAL TOKENS VS. BASELINE	1.000	0.661	0.950	0.659	1.000
DATASET EMBEDDINGS VS. BASELINE	0.996	0.995	1.000	0.996	0.995

Table 4: ASO scores of the special tokens model and dataset embedding model compared to the baseline

C Group Contributions

All members contributed evenly to the project.